

SYSTEMS ENGINEERING

Systems Analysis & Design

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Outline

- 1 Basic Concepts
- 2 Systems Engineering
- 3 Cybernetics and Technology
- 4 Teams-Based Structure as a System



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What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance** of **complex systems**.
- This **discipline** is based on interdisciplinary fields such as control engineering, industrial engineering, software engineering, mechanical engineering, electrical engineering, organizational studies, project management, and others.
- **Systems Engineering** is a **holistic** approach to engineering that focuses on how to **design** and manage **complex systems** over their **lifecycle**.



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Systems Engineer as a Professional

- A **Systems Engineer** is a **professional** who is responsible for **designing, implementing, and maintaining complex systems**.
- A **Systems Engineer** must have a broad understanding of engineering, mathematics, science, and technology.
- A **Systems Engineer** must be able to analyze and solve problems in a systematic and holistic way.



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Human Activities

- **Human activities** are **complex systems** that involve **multiple components, interactions, and feedback loops**.
- **Systems Engineering** can be applied to **understand, analyze, and improve human activities**.
- **Systems Engineering** can help us **design and manage complex systems** such as **organizations, cities, economies, and societies**.
- **Interactions** are **key to understanding and improving human activities**.
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Organizations as Systems

- **Organizations** can be **viewed** as: a **rational system**, a **natural system**, or an **open system**.
- A **rational system** is a formal organization that is designed to achieve specific goals as a machine.
- A **natural system** is an informal organization that is emergent and adaptive based on human interactions.
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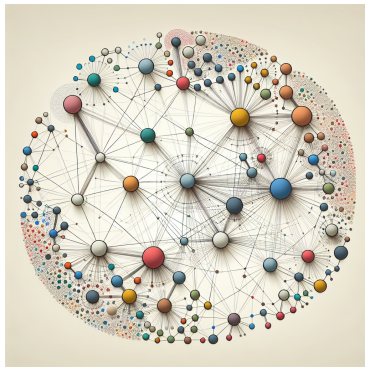
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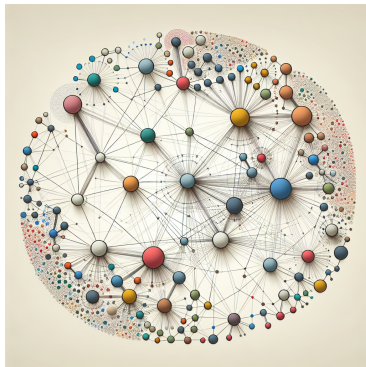
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- It means the **interactions** can **boost the capabilities** of the parts of the **system**. Also, it allows both understanding **emergent behaviors** and defining **improvements** in **systems**.
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Real Madrid Galácticos: Systems Engineering Case Study

- **Galácticos Era** (2000 — 2006): Real Madrid assembled the most **expensive** and **talented** individual players in football history.
- **Star Players**: Zinedine Zidane, Robinho, David Beckham, Luís Figo, Ronaldo, Kaká. Roberto Carlos, ... each a **world-class** individual performer.
- **Expected Result**: With the best **components**, success should be **guaranteed**, right? Watch this video.
- **Real Result**: Only **one** Champions League in six years, **two** La Liga titles, but multiple early Champions League exits. Poor **team chemistry** and lack of **system integration** despite massive investment.
- **Systems Lesson**: A collection of excellent *parts* does not automatically create an excellent *system*. **Synergy** and **integration** matter more than individual talent.

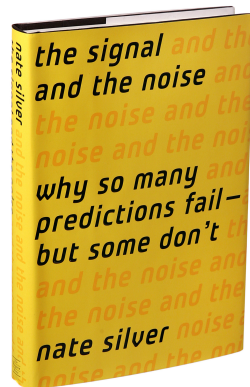


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Synergy: Money Ball



Talking with Machines!

- **Alan Turing** proposed a century ago a **Universal Machine**, capable of taking any **algorithm** defined as a **state machine**, and processing it in a **binary language**.
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Programming Languages

- **Programming Languages** with more **capabilities** and easier **comprehension** have been created. Also, more people started to **code** in specific **domain programming languages**.

Demo time!

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Applications of Systems Engineering

- **Systems Engineering** can be applied to **understand**, **analyze**, and **improve complex systems**.
- **Systems Engineering** can be applied to **design** and **manage complex systems** such as **organizations**, **cities**, **economies**, and **societies**.
- **Systems Engineering** can be applied to **design** and **manage complex systems** such as **software**, **hardware**, and **networks**.



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Cybernetics

- **Cybernetics** is the **study** of **systems**, **control**, and **communication** in *animals, machines, and organizations*.
- **Cybernetics** is a transdisciplinary field that **combines** **engineering**, **mathematics**, **biology**, **psychology**, and **philosophy**.
- **Cybernetics** is the foundation of **systems engineering** and **information technology**.



Technology

- **Technology** is the *application* of scientific knowledge to **solve problems** and **improve systems**.
- **Technology** is the *key* to **designing** and **managing complex systems** such as organizations, cities, economies, and societies.



Technology: AI

- **Artificial Intelligence** is a *field* of computer science that *studies* how to **design** and **implement intelligent agents**.
- Artificial Intelligence is the key to *designing* and *managing* **complex systems** such as smart-organizations, smart-cities, smart-economies, and smart-societies.

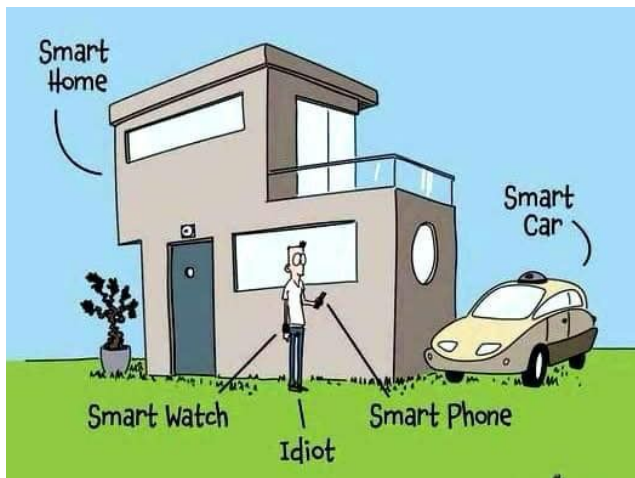


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Artificial Intelligence **not** as a System



Cybernetics & Technology in Systems Context

- **Cybernetics** and **technology** are the foundation of systems engineering and information technology.
- **Cybernetics** and **technology** are the foundation of artificial intelligence and smart systems.



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Team-Based Structure Organizations

- **Team-based structure organizations** are a way to **organize work** and **people** in **teams** that are self-managed and cross-functional.
- Each team is **responsible** for a specific task or project and has the **authority** to **make decisions** and **solve problems**.
- **Team-based structure organizations** are flexible, agile, and innovative because they **empower employees** and **encourage collaboration**.



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Teams as a System

- **Teams** are a **system** where collaboration (synergy) and communication are **key** to **success**.
- Defining and automating the **processes** and **procedures** within the **team** is a **challenge** to **improve team performance**.
- **Teams** are like **pieces** in a **puzzle**, where each **piece** has a **specific role** and **responsibility**.



"Soft Skills" in Engineering

- **Soft skills** are **personal attributes** that enable someone to **interact** effectively and harmoniously with other **people**.
- Typical Soft Skills:
 - **Communication** skills (verbal and written).
 - **Teamwork** and collaboration.
 - **Problem-solving** and critical thinking.
 - **Adaptability** and flexibility.
 - **Time management** and organization.
 - **Leadership** and management.
 - **Emotional intelligence**.
 - **Creativity** and innovation.
 - **Conflict resolution**.
 - **Networking** and relationship building.
 - **Customer service** and client management.



Computer Analyst — Technical Skills & Responsibilities

● Skills:

- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

● Responsibilities:

- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, such as functional specifications and use cases.
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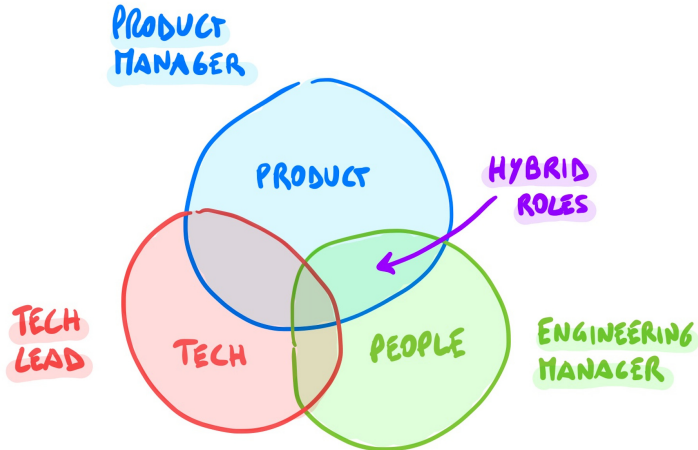
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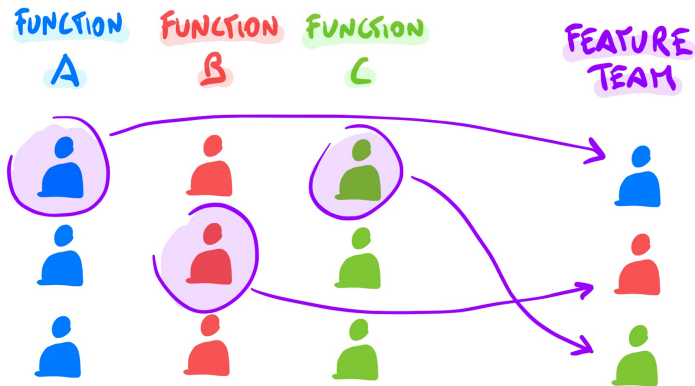
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Technical Team Typical Structure



Feature Teams



What does it mean to be a leader? I

- **Leading** a team is **not a role**. It is a decision; you can be a **leader** anytime and anywhere.
- **Teamwork culture** is very **important**. It creates habits, open communication, and safe spaces for inclusion.
- **Psychological safety** is a **key** point for having an **effective** team. You can develop *technical skills*, but **it is not enough**.
- **Hierarchy** is very important. **Anarchism** tends to fail. **Hierarchy** exists through **status** and **power**.
- In a **hierarchy**, **experts lead** to make **better decisions**. However, everyone must be **careful** not to leave **people** behind.



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- With **crystal-clear communication** and clarity about **business goals** and achievements, people feel **more comfortable** pursuing the same **goals** as a **team**.
- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just completing tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish—just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed over time**. Either way, **context** and **the desire** for **self-growth** are **vital**.
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- A **good leader** builds **trusting relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To develop as a leader, follow the **three C's**: **Curiosity**, **Courage**, and **Commitment**.



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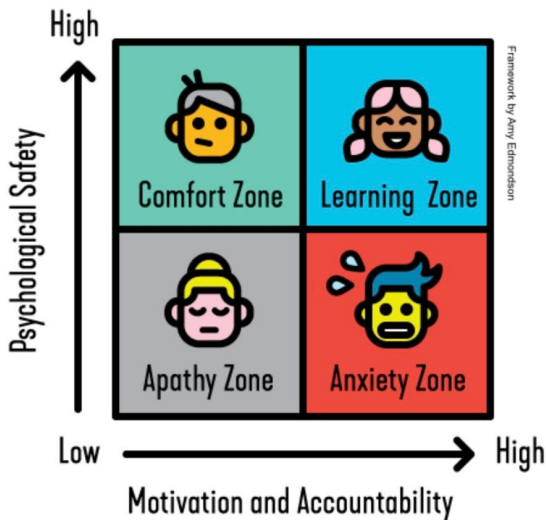


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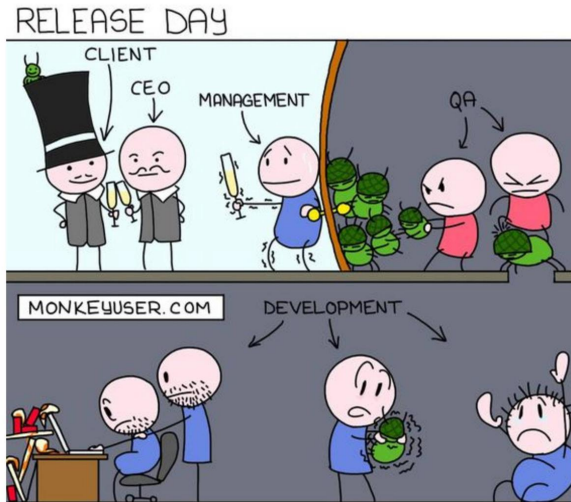
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Working Zones



Real World!



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Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

